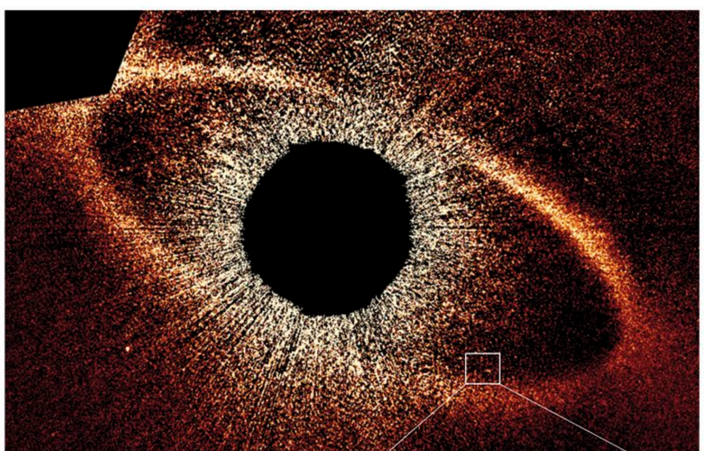
EXTRASOLAR PLANETARY SYSTEMS

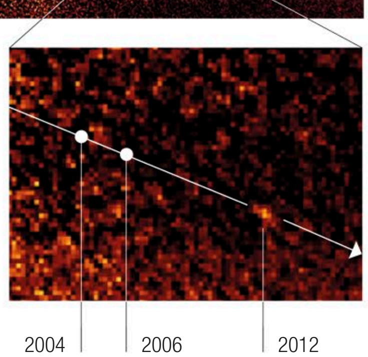
ANY GROUP OF PLANETS ORBITING A STAR OTHER THAN THE SUN IS CALLED AN EXTRASOLAR PLANETARY SYSTEM. THE INDIVIDUAL PLANETS CIRCLING AROUND IN THESE SYSTEMS ARE CALLED EXOPLANETS.

More than 2,000 exoplanets have been discovered so far, mostly in the last ten years or so. About half are gas-dominated planets, about the size of Jupiter or Neptune in the Solar System, orbiting close to their host stars. These hellishly hot, star-snuggling gas giants are known as "hot Jupiters" or "hot Neptunes." Many smaller, probably rocky, exoplanets have also been discovered—some about the size of Earth—as well as cold gas giants. The types of stars that exoplanets orbit vary from red dwarfs to Sun-like stars, red giants, and even pulsars.

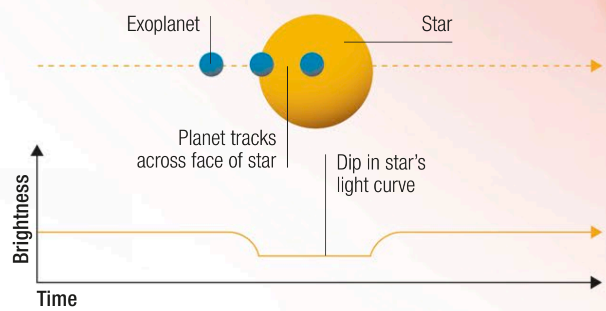
Perhaps the most remarkable fact about exoplanets is that they can be detected at all. Finding a body many light-years away that emits no light of its own and that orbits a much bigger, brighter body (a star) presents many challenges. So far, relatively few exoplanets have been imaged directly with telescopes, but around a dozen methods have been devised for detecting them indirectly. Three of the most successful of these methods are explained below.

On average, **each star** in the Milky Way galaxy has **at least one planet** orbiting it

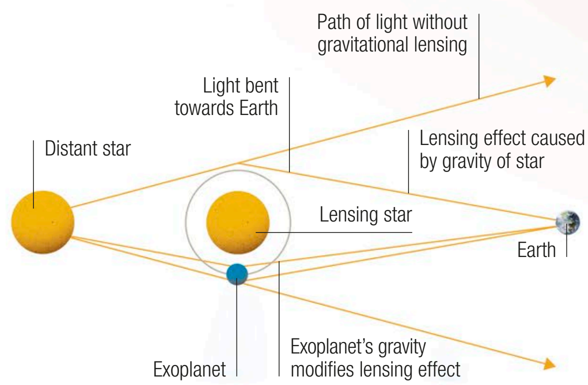




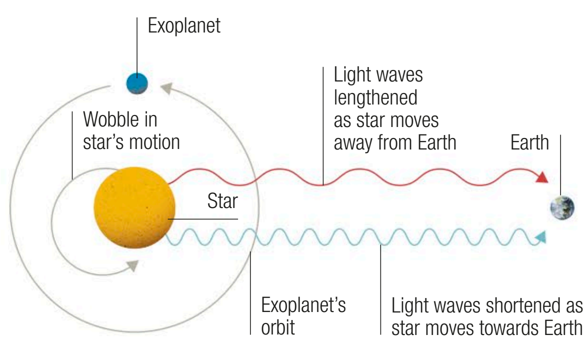
△ **Direct imaging**
The star Fomalhaut has a disk of dust and gas around it, as shown above (the star itself has been blacked out). A planet in the disk has been directly imaged by the Hubble Space Telescope. The planet and its path are shown in the image to the right.



▷ **Transit method**
This approach involves detecting miniscule dips in a star's brightness, caused by transits (movements) of a planet across the face of the star. To do this, an extremely sensitive light-detecting instrument is used.



▷ **Gravitational microlensing**
The gravity of a star can bend light coming from a more distant star. This means it can act like a lens and magnify the distant star as it appears from Earth. An exoplanet orbiting the lensing star produces detectable variations in the amount of magnification.



▷ **Doppler spectroscopy**
An exoplanet's orbit causes a "wobble" in the motion of its host star. As a result, light waves coming from the star are alternately slightly lengthened (making them look redder) and shortened (making them look bluer)—a measurable phenomenon.